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The Suburban Street Design That's Been Driving Emissions Since WWII

A century-old planning paradigm still embedded in the physical fabric of suburbs nationwide is driving a significant share of greenhouse gas emissions.

Half of all Americans live in the suburbs. For decades, planners and policymakers have blamed suburban sprawl's environmental and social costs on one thing: distance. The farther people live from city centers, the more they drive, the more carbon they emit, and the more disconnected

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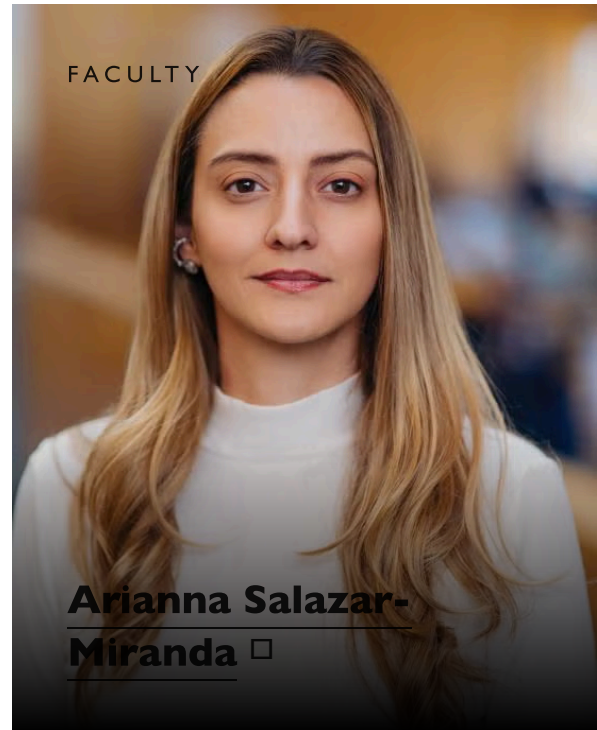
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has received.

In the height of 20th century suburban growth in the U.S., planners relied heavily on Garden City Design (GCD) to provide peaceful, picturesque, and aesthetically pleasing neighborhoods with windy roads and cul-de-sacs removed from urban noise and congestion. Together, these features structurally enforce car dependency, the study published in [*Nature Sustainability*](#) found.



Suburbanization as a whole adds 0.26 metric tons of greenhouse gas emissions per person per year. GCD accounts for 0.10 metric tons, or about 38%, of the environmental costs of suburbs.

“The paper shows that a substantial share of the costs we attribute to ‘sprawl’ actually stem from street design,” Salazar-Miranda said. “Right now, the conversation is almost entirely about distance to downtown, and that matters, but an important and overlooked part of the costs comes from how neighborhoods were designed. The winding streets and cul-de-sacs turn what could be short, direct trips for grocery runs, errands, and recreation, into longer ones.”

GCD originated in early 20th-century Britain and became the blueprint for American suburban development after World War II. To measure its reach, Salazar-Miranda constructed a composite index drawing of its main design features and applied it to more than

- Curvilinear streets that increase distance between two points.
- Cul-de-sacs and dead-end streets which eliminate through-routes and force residents onto arterial roads that add mileage.
- Hierarchical road networks that funnel all traffic onto major arteries and make walking or biking impractical and dangerous.
- Irregular block layouts that restrict where commercial activity can locate and reduce shops, services, and amenities within walking distance.

Since design accounts for a meaningful share of the harm, targeted retrofitting of existing suburbs — reconnecting street networks, adding sidewalks, introducing mixed-use zoning — is a legitimate and potentially cost-effective policy lever, the study argues.

“The good news is that design is something that we can change, even in neighborhoods that were built decades ago. Several U.S. cities are already doing this.

Boston and San Francisco have removed highways to reconnect neighborhoods. Portland requires frequent street connections in new developments. Virginia restricts cul-de-sac subdivisions unless they meet connectivity standards,” Salazar-Miranda said. “We can improve outcomes in suburbs that already exist and for new suburbs being built right now. We don't have to repeat the design mistakes of the 20th century.”

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